Why the Interference Indices Are Interleaved: Tracing Every γ-Matrix Back to Its Vertex

PHYS 7502

1 The Source of Every γ-Matrix

Every γ-matrix in the trace comes from a specific vertex in a specific diagram. Each vertex connects to a specific photon, which assigns its Lorentz index. Let us label each γ by its origin.

The Two Diagrams' Vertex Structure

$\mathcal{M}_s$ **(s-channel):** The electron absorbs $\gamma_{\text{in}}(p_2)$ at Vertex A, then emits $\gamma_{\text{out}}(p_3)$ at Vertex B.
Reading backward along the fermion line ($\bar{u}(4) \leftarrow \cdots \leftarrow u(1)$):

$$\bar{u}(4) \quad \underbrace{\gamma^{\alpha}}_{\substack{\text{Vertex B}\\ \text{emits photon 3}\\ \text{index from } \varepsilon_3^{*\alpha}}} \quad (\not{p}_1 + \not{p}_2) \quad \underbrace{\gamma^{\beta}}_{\substack{\text{Vertex A}\\ \text{absorbs photon 2}\\ \text{index from } \varepsilon_2^{\beta}}} \quad u(1)$$

$\mathcal{M}_t$ **(t-channel):** The electron emits $\gamma_{\text{out}}(p_3)$ at Vertex A, then absorbs $\gamma_{\text{in}}(p_2)$ at Vertex B.
Reading backward along the fermion line:

$$\bar{u}(4) \quad \underbrace{\gamma^{\mu}}_{\substack{\text{Vertex B}\\ \text{absorbs photon 2}\\ \text{index from } \varepsilon_2^{\mu}}} \quad (\not{p}_1 - \not{p}_3) \quad \underbrace{\gamma^{\nu}}_{\substack{\text{Vertex A}\\ \text{emits photon 3}\\ \text{index from } \varepsilon_3^{*\nu}}} \quad u(1)$$

Key difference: The photon-vertex assignments are *swapped* between the two diagrams. In $\mathcal{M}_s$, photon 3 is at the left γ and photon 2 at the right γ. In $\mathcal{M}_t$, it is reversed.

2 Conjugating $\mathcal{M}_t$ to Get $\mathcal{M}_t^*$

Under conjugation, the spinors swap, the γ-matrices reverse order, and polarizations toggle their stars:

$$\mathcal{M}_t^* \propto \bar{u}(1) \quad \underbrace{\gamma^{\nu}}_{\substack{\text{was rightmost}\\ \text{now leftmost}}} \quad (\not{p}_1 - \not{p}_3) \quad \underbrace{\gamma^{\mu}}_{\substack{\text{was leftmost}\\ \text{now rightmost}}} \quad u(4)$$

with $\varepsilon_2^{\mu} \to \varepsilon_2^{*\mu}$ and $\varepsilon_3^{*\nu} \to \varepsilon_3^{\nu}$.

What Conjugation Did to the Index Positions

In $\mathcal{M}_t$, the reading order was: μ (photon 2) then ν (photon 3).
In $\mathcal{M}_t^*$, the reversed reading order is: ν (photon 3) then μ (photon 2).
The **photon identity** of each γ does not change under conjugation — a γ that came from the photon-2 vertex still carries a photon-2 index. Only the **position** in the chain reverses.

3 Building $|\mathcal{M}_s|^2$ vs. $\mathcal{M}_s\mathcal{M}_t^*$: The Key Comparison

Casimir's trick stitches the two bilinears into a trace by inserting $\not{p}_1$ (from particle 1 completeness) between the right end of the first bilinear and the left end of the second, and $\not{p}_4$ (from particle 4 completeness) to close the loop. The trace reads:

$$\mathrm{Tr}\Big[\underbrace{\text{guts of first factor}}_{\Gamma_1}\not{p}_1\underbrace{\text{guts of second factor}}_{\Gamma_2}\not{p}_4\Big]$$

Let us now see what Γ_1 and Γ_2 are in each case.

3.1 Case 1: $|\mathcal{M}_s|^2 = \mathcal{M}_s \times \mathcal{M}_s^*$

$|\mathcal{M}_s|^2$: Where Each γ Comes From

Γ_1 comes from $\mathcal{M}_s$:

$$\Gamma_1 = \underbrace{\gamma^\alpha}_{\substack{s\text{-channel}\\ \text{Vertex B}\\ \text{photon 3}}}(\not{p}_1+\not{p}_2)\underbrace{\gamma^\beta}_{\substack{s\text{-channel}\\ \text{Vertex A}\\ \text{photon 2}}}$$

Γ_2 comes from $\mathcal{M}_s^*$ (reversed order, new index names):

$$\Gamma_2 = \underbrace{\gamma^\sigma}_{\substack{s^*\text{-channel}\\ \text{Vertex A}\\ \text{photon 2}}}(\not{p}_1+\not{p}_2)\underbrace{\gamma^\rho}_{\substack{s^*\text{-channel}\\ \text{Vertex B}\\ \text{photon 3}}}$$

The full trace:

$$\mathrm{Tr}\Big[\underbrace{\gamma^\alpha}_{\text{ph3}}(\not{p}_1+\not{p}_2)\underbrace{\gamma^\beta}_{\text{ph2}}\not{p}_1\underbrace{\gamma^\sigma}_{\text{ph2}}(\not{p}_1+\not{p}_2)\underbrace{\gamma^\rho}_{\text{ph3}}\not{p}_4\Big]$$

After photon sums (α contracts with ρ, β contracts with σ):

$$\mathrm{Tr}\Big[\gamma^\alpha\cdots\gamma^\beta\cdots\gamma_\beta\cdots\gamma_\alpha\cdots\Big]$$

Pattern: NESTED. The β pair sits *inside* the α pair:

$$\alpha\cdots\beta\cdots\beta\cdots\alpha$$

3.2 Case 2: $\mathcal{M}_s \times \mathcal{M}_t^*$

$\mathcal{M}_s\mathcal{M}_t^*$: Where Each γ Comes From

Γ_1 comes from $\mathcal{M}_s$ (same as before):

$$\Gamma_1 = \underbrace{\gamma^\alpha}_{\substack{s\text{-channel}\\ \text{Vertex B}\\ \text{photon 3}}} (\not{p}_1 + \not{p}_2) \underbrace{\gamma^\beta}_{\substack{s\text{-channel}\\ \text{Vertex A}\\ \text{photon 2}}}$$

Γ_2 comes from $\mathcal{M}_t^*$ (the ***t*-channel** conjugate):

$$\Gamma_2 = \underbrace{\gamma^\nu}_{\substack{t^*\text{-channel}\\ \text{Vertex A}\\ \text{photon 3}}} (\not{p}_1 - \not{p}_3) \underbrace{\gamma^\mu}_{\substack{t^*\text{-channel}\\ \text{Vertex B}\\ \text{photon 2}}}$$

The full trace:

$$\mathrm{Tr}\Big[\underbrace{\gamma^\alpha}_{\text{ph3}} (\not{p}_1 + \not{p}_2) \underbrace{\gamma^\beta}_{\text{ph2}} \not{p}_1 \underbrace{\gamma^\nu}_{\text{ph3}} (\not{p}_1 - \not{p}_3) \underbrace{\gamma^\mu}_{\text{ph2}} \not{p}_4 \Big]$$

After photon sums (α contracts with ν, β contracts with μ):

$$\mathrm{Tr}\Big[\gamma^\alpha \cdots \gamma^\beta \cdots \gamma_\alpha \cdots \gamma_\beta \cdots \Big]$$

Pattern: INTERLEAVED. The pairs alternate:

$$\alpha \cdots \beta \cdots \alpha \cdots \beta$$

4 Why the Patterns Are Different

The Root Cause

The difference comes from the **photon-vertex assignment being swapped** between the s- and t-channel diagrams.

In $\mathcal{M}_s$: the γ-matrix order (reading backward) is photon 3, then photon 2:

$$\bar{u}(4) \underbrace{\gamma}_{\text{ph3}} \cdots \underbrace{\gamma}_{\text{ph2}} u(1)$$

In $\mathcal{M}_t^*$: the conjugated t-channel has the *same* order (photon 3, then photon 2):

$$\bar{u}(1) \underbrace{\gamma}_{\text{ph3}} \cdots \underbrace{\gamma}_{\text{ph2}} u(4)$$

But in $\mathcal{M}_s^*$: the conjugated s-channel reverses to (photon 2, then photon 3):

$$\bar{u}(1) \underbrace{\gamma}_{\text{ph2}} \cdots \underbrace{\gamma}_{\text{ph3}} u(4)$$

So when you stitch Γ_1 (from $\mathcal{M}_s$) with Γ_2 (from the conjugate):

If Γ_2 is from $\mathcal{M}_s^*$: photon order in Γ_2 is (ph2, ph3) — *reversed* from Γ_1 (ph3, ph2). Same-photon γ's face each other across the stitch → **nested**.
If Γ_2 is from $\mathcal{M}_t^*$: photon order in Γ_2 is (ph3, ph2) — *same* as Γ_1 (ph3, ph2). Same-photon γ's are on the same side → **interleaved**.

5 Seeing It Visually

The Trace as a Loop

Think of the trace as a closed loop. Label each position with its photon identity:

$|\mathcal{M}_s|^2$ **(nested):**

$$\overbrace{\gamma^{\alpha}}^{\substack{\text{pos 1}\\ \text{ph3}}} - (\not{p}_1 + \not{p}_2) - \overbrace{\gamma^{\beta}}^{\substack{\text{pos 3}\\ \text{ph2}}} - \not{p}_1 - \overbrace{\gamma_{\beta}}^{\substack{\text{pos 5}\\ \text{ph2}}} - (\not{p}_1 + \not{p}_2) - \overbrace{\gamma_{\alpha}}^{\substack{\text{pos 7}\\ \text{ph3}}} - \not{p}_4$$

The blue pair (ph2) is positions 3 and 5 — **adjacent, inner**.
The red pair (ph3) is positions 1 and 7 — **wrapping around, outer**.
⇒ You can contract the blue pair *first* without disturbing the red pair.

$\mathcal{M}_s\mathcal{M}_t^*$ **(interleaved):**

$$\overbrace{\gamma^{\alpha}}^{\substack{\text{pos 1}\\ \text{ph3}}} - (\not{p}_1 + \not{p}_2) - \overbrace{\gamma^{\beta}}^{\substack{\text{pos 3}\\ \text{ph2}}} - \not{p}_1 - \overbrace{\gamma_{\alpha}}^{\substack{\text{pos 5}\\ \text{ph3}}} - (\not{p}_1 - \not{p}_3) - \overbrace{\gamma_{\beta}}^{\substack{\text{pos 7}\\ \text{ph2}}} - \not{p}_4$$

The red pair (ph3) is positions 1 and 5.
The blue pair (ph2) is positions 3 and 7.
They are **interleaved**: red-blue-red-blue. You *cannot* contract one pair without the other being in the way. That is why you need Theorem 9 (which handles 3 matrices between a contracted pair) before you can use Theorem 8′.

6 Summary: Why the Indices End Up Where They Do

The Logic Chain

1. Each QED vertex assigns a γ-matrix index from its photon's polarization vector.
2. The s-channel and t-channel **swap which photon goes to which vertex**.
3. Conjugation **reverses** the γ-matrix order within each amplitude.
4. Casimir's trick **stitches** Γ_1 (from the unconjugated amplitude) and Γ_2 (from the conjugated amplitude) into a single trace, separated by $\not{p}_1$ and $\not{p}_4$.

5. For $|\mathcal{M}_s|^2$: Γ_1 and Γ_2 both come from the s-channel. Conjugation reverses the photon order in Γ_2, so same-photon γ's end up facing each other $\rightarrow$ **nested**.

6. For $\mathcal{M}_s\mathcal{M}_t^*$: Γ_1 comes from the s-channel, Γ_2 from the t-channel. The t-channel already has swapped photon order, and conjugation reverses it back to the same order as Γ_1. Same-photon γ's end up on the same side $\rightarrow$ **interleaved**.

7. The interleaved pattern requires Theorem 9 to untangle, which ultimately produces the factor $(s + t + u) = 0$.